Name: ____________

Date: ____________

Sequences & Series

Student learning objectives for this unit

SUPPORTING SKILLS LEGEND

I — Introduce | D — Deepen | A — Apply | R — Reinforce

- [ ] **1. I can evaluate and graph a sequence defined explicitly or recursively.**

RELATED SKILLS

- I — Generate a recursive formula with an initial condition.
- I — Generate an explicit formula from a sequence pattern.
- I — Graph a sequence as discrete ordered pairs.
- I — Interpret subscripts and evaluate terms of a sequence.
- I — Recall and correctly use the terms sequence, term, index, explicit formula, recursive formula, arithmetic sequence, and geometric sequence.

- [ ] **2. I can write explicit and recursive formulas for arithmetic sequences.**

RELATED SKILLS

- I — Recognize constant difference in an arithmetic sequence.
- D — Generate a recursive formula with an initial condition.
- D — Generate an explicit formula from a sequence pattern.

- [ ] **3. I can write explicit and recursive formulas for geometric sequences.**

RELATED SKILLS

- I — Recognize constant ratio in a geometric sequence.
- D — Generate a recursive formula with an initial condition.
- D — Generate an explicit formula from a sequence pattern.

- [ ] **4. I can distinguish arithmetic, geometric, and other sequences from formulas, tables, or contexts.**

RELATED SKILLS

- D — Recognize constant difference in an arithmetic sequence.
- D — Recognize constant ratio in a geometric sequence.
- A — Compare functions shown in different representations.

- [ ] **5. I can create and interpret a sequence model for discrete change.**

RELATED SKILLS

- I — Identify initial value and pattern of change in a discrete context.
- A — Explain a model parameter in the language of its context.
- A — Generate a recursive formula with an initial condition.
- A — Generate an explicit formula from a sequence pattern.

- [ ] **6. I can interpret and use summation notation.**

RELATED SKILLS

- I Expand and evaluate a finite sum written in sigma notation.
- I Recall and correctly use the terms series, partial sum, sigma notation, finite series, infinite series, and convergence.
- A Interpret subscripts and evaluate terms of a sequence.

- [ ] **7. I can calculate a finite arithmetic series.**

RELATED SKILLS

- I Apply an arithmetic-series sum formula.
- A Recognize constant difference in an arithmetic sequence.

- [ ] **8. I can calculate a finite geometric series.**

RELATED SKILLS

- I Apply a finite geometric-series sum formula.
- A Recognize constant ratio in a geometric sequence.

- [ ] **9. I can determine whether an infinite geometric series converges and find its sum when it does.**

RELATED SKILLS

- I Test convergence and sum an infinite geometric series.
- A Recognize constant ratio in a geometric sequence.

- [ ] **10. I can create and interpret a series model for accumulated change.**

RELATED SKILLS

- I Identify the initial term, change factor, and number of terms in an application.
- A Apply a finite geometric-series sum formula.
- A Apply an arithmetic-series sum formula.
- A Explain a model parameter in the language of its context.
- A Test convergence and sum an infinite geometric series.
